



IILM
UNIVERSITY

PROMPT ENGINEERING AND GENERATIVE AI

Internship Project & Learning Outcomes

Name: Shashank Raj

Enrollment No: 25SCS1003001227

Program & Semester: B. Tech CSE • Batch 2025–29 • 2CSE_14

Organization: InAmigos Foundation

2025–29

Internship Overview

01

1

Core Focus

Practical learning in prompt engineering and Generative AI through structured activities.

2

Learning Approach

Create → test → observe → compare → refine
→ document.

3

Practical Context

AI-assisted planning, learning, communication, decision-making, interviews and project organization.

- Four supplied activities formed the internship scope: Prompt Creation, Prompt Improvement, Prompt Role-Play and Prompt Workflow.
- Central insight: clearer instructions produce more relevant, structured and actionable AI responses.
- Responsible AI use emphasized verification, independent thinking and avoiding blind copying.

Organization & Internship Context

02

InAmigos Foundation

The report places the internship in the context of an educational and skill-development programme associated with InAmigos Foundation.

Nature of the Work

The supplied activities emphasize practical AI literacy, experimentation, comparison of approaches and documentation of learning.

- The internship focused on prompt-engineering skills rather than development of a production AI model.
- Activities applied Generative AI to education, planning, communication, career preparation, decision-making and project organization.
- The report deliberately avoids unsupported statutory or institutional claims about the organization.

Problem Statement & Objectives

03

Problem Statement

Vague AI requests can produce technically reasonable answers that do not match the user's actual purpose. The internship explored how structure changes response quality.

- Understand how prompt wording affects relevance, quality and usefulness.
- Construct prompts using role, context, objective, audience, constraints, examples and output format.
- Compare basic and improved prompts and identify the techniques responsible for better outputs.
- Study role assignment and its effect on tone, priorities and decision-making.
- Build connected multi-step prompt workflows.
- Develop responsible AI-use habits, including verification and institutional compliance.

Project Scope & Major Tasks

04

1

Prompt Creation

10 prompts covering planning, learning, problem-solving, decision-making, summarization, ideas, writing, critical thinking, interviews and goal-setting.

2

Prompt Improvement

5 basic prompts refined by adding context, audience, constraints, structure, measurable tasks and evaluation.

3

Prompt Role-Play

5 professional roles: Teacher, HR Professional, Event Planner, Career Advisor and Project Manager.

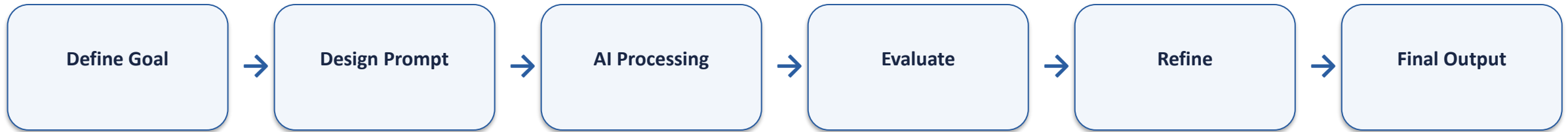
4

Prompt Workflow

4 connected stages for Software Developer interview preparation.

Methodology & System Architecture

05



- Experiment-and-refine methodology: each activity began with a practical goal, followed by prompt construction, AI testing, observation, comparison and documentation.
- The same cycle was applied across individual prompts, role-based prompts and connected workflows.
- Evaluation focused on relevance, clarity, completeness, structure and usefulness.

Task 1 — Prompt Creation

06

Area	Example	Main Technique
Planning	Study schedule	Context + constraints + format
Learning	Embeddings in Generative AI	Role + simple language + analogy
Problem-solving	Aptitude problem	Step-by-step + verification
Decision-making	₹3,000 choice	Criteria + scoring
Summarization	AI in everyday life	Word limit + structure
Idea generation	Technology projects	Constraints + ranking
Writing	AI-use paragraph	Audience + tone + word limit
Critical thinking	AI in academic assignments	Pros + cons + conclusion
Interview	Internship mock interview	Role + interaction rules
Goal-setting	30-day Java plan	Time limit + stages

Task 2 — Prompt Improvement

07

Area	Basic Prompt	Improvement Technique
Learning a concept	“Explain machine learning.”	Audience + structure + constraints + examples
Study planning	“Make a study plan for my exams.”	Context + time constraints + priorities + format
Summarization	“Summarize Generative AI.”	Purpose + word limit + required content + tone
Programming skills	“How can I improve my programming?”	Specific goal + progression + measurable tasks + fallback
Decision-making	“Which skill should I learn first?”	Context + evaluation criteria + scoring + recommendation

Overall Finding

Improved prompts were more immediately usable, consistent and aligned with the intended purpose.

Task 3 — Prompt Role-Play

08

Role	Tone / Focus	Response Style
Teacher	Educational; understanding concepts	Explanations, examples, questions
HR Professional	Professional; employability & behavior	Interview questions, sample answers, HR reasoning
Event Planner	Practical; execution & logistics	Schedule, resources, engagement, risks
Career Advisor	Motivational; long-term development	Roadmap, milestones, measurable outcomes
Project Manager	Structured; delivery & accountability	Scope, roles, deadlines, dependencies, risks, testing

- Strongest example in the report: Project Manager, because the role naturally supports planning, task allocation, risk management, deadlines, testing and accountability.
- Key finding: role + context + clear instructions produces a more focused result than a broad request.

Task 4 — Prompt Workflow

09

1 Understand the Job Role

Identify technical skills, soft skills, responsibilities and preparation topics.

2 Generate Interview Questions

Create 8 technical, 4 HR/behavioural and 3 problem-solving questions.

3 Practice & Improve Answers

Ask one at a time; score answers on five criteria and provide targeted feedback.

4 Final Preparation Plan

Turn role analysis and feedback into a seven-day plan and interview-day checklist.

Technologies & Techniques

10

1

Generative AI / AI tool

Creation, testing, refinement, comparison, explanation, planning and mock-interview activities.

2

Microsoft Word

Preparation and organization of task documents and the final internship report.

3

Prompt-engineering techniques

Role assignment, context, constraints, output formats, examples, evaluation criteria, interaction and verification.

4

Tables & structured formats

Comparison of responses, schedules, decision matrices, roadmaps, checklists and workflow stages.

Learning Outcomes & Key Findings

- Improved ability to write clear, specific and goal-oriented prompts.
- Better understanding of how audience, context, constraints and output format influence AI responses.
- Ability to compare vague and structured prompts and explain why their outputs differ.
- Understanding of role-based prompting and how professional perspectives change priorities.
- Ability to design connected prompt workflows rather than treating AI as only a one-question/one-answer tool.
- Improved planning, communication, interview preparation and project-organization skills.
- Greater awareness of verification, independent thinking, responsible AI use and academic integrity.

Key Takeaway

“AI output becomes more useful when prompts provide clear role, context, objectives, constraints and evaluation.”

Challenges & Responsible AI Practice

- Distinguishing useful AI output from generic or poorly aligned responses.
- Refining prompts when important context, constraints or evaluation criteria are missing.
- Keeping answers aligned with audience, purpose and required format.
- Connecting multiple prompt stages without losing continuity.
- Verifying AI-generated content and avoiding blind copying.

Responsible Use

The internship emphasized verification, independent thinking, following institutional policies and using AI as an assistant rather than a substitute for judgment.

Overall Internship Outcome

13

What I Completed

- 10 prompt-creation examples
- 5 prompt-improvement comparisons
- 5 professional role-play perspectives
- 1 four-step interview-preparation workflow
- Structured documentation of findings

What I Gained

- Prompt-design skills
- Analytical comparison skills
- Practical Generative AI literacy
- Planning and communication skills
- Interview-preparation skills
- Responsible AI-use habits

The internship converted prompt engineering from a theoretical concept into a repeatable practical skill.

THANK YOU

PROMPT ENGINEERING AND GENERATIVE AI

Shashank Raj • 25SCS1003001227 • B.Tech CSE